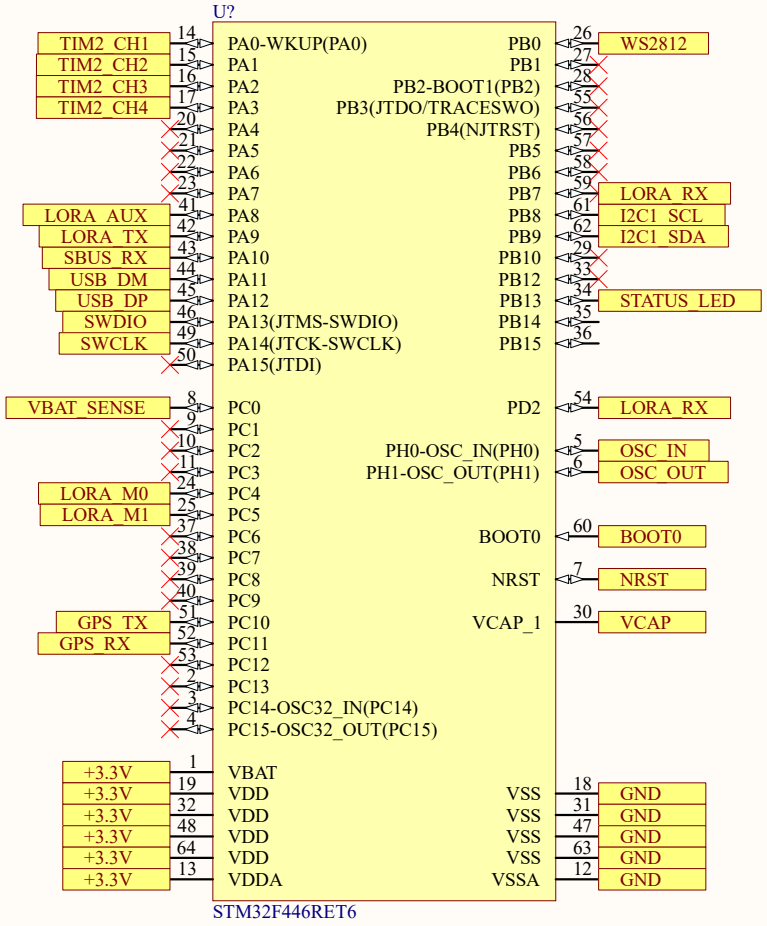
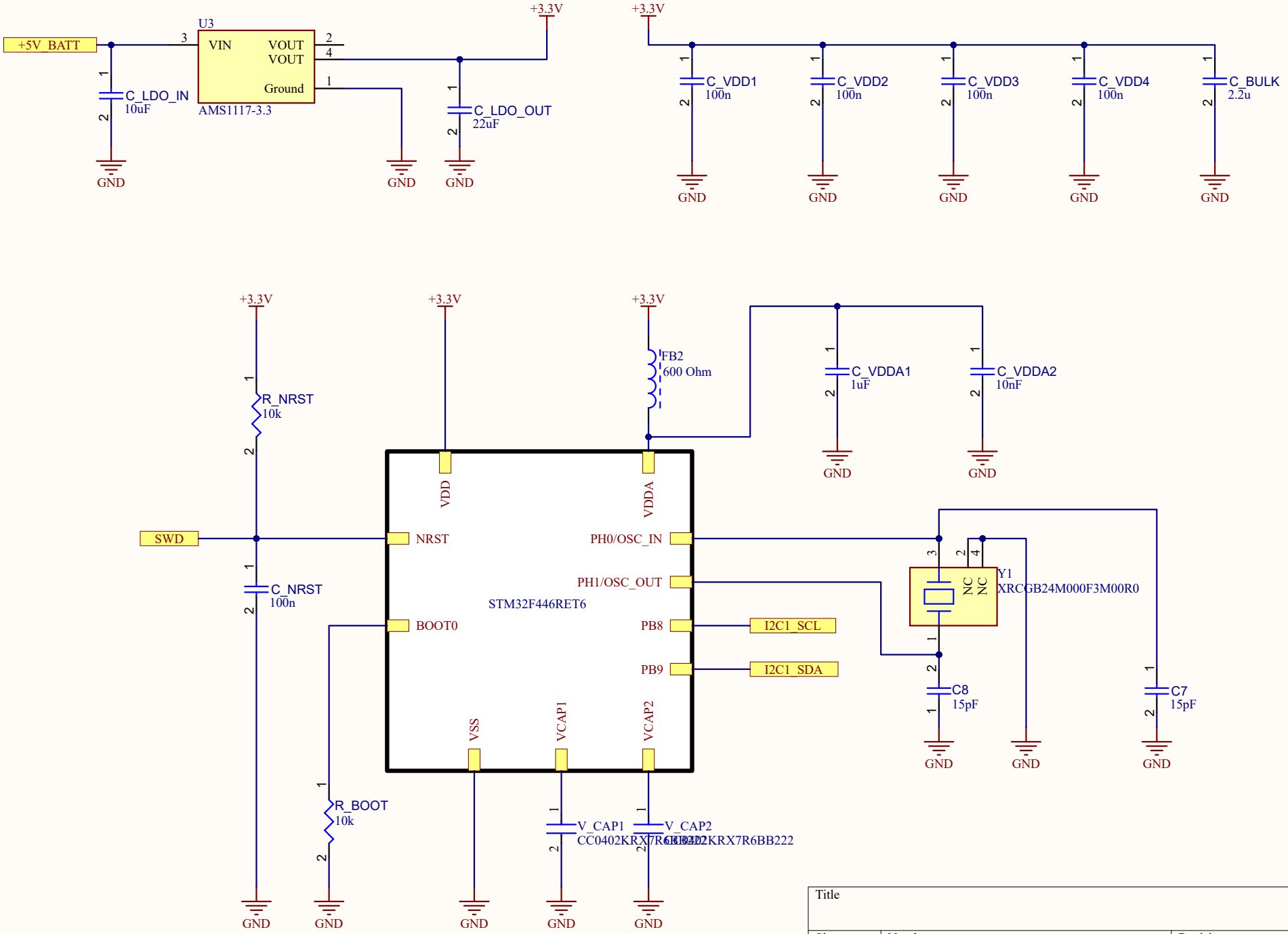
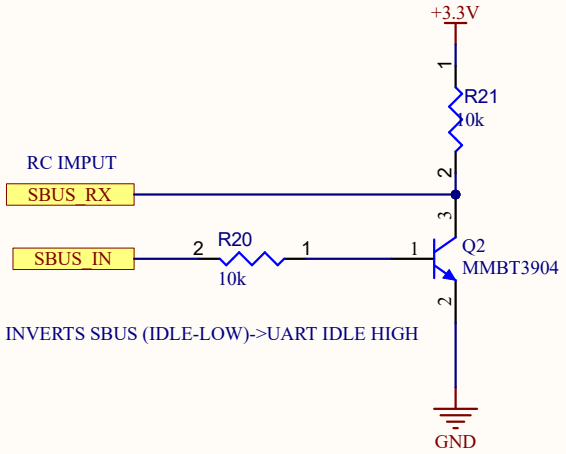


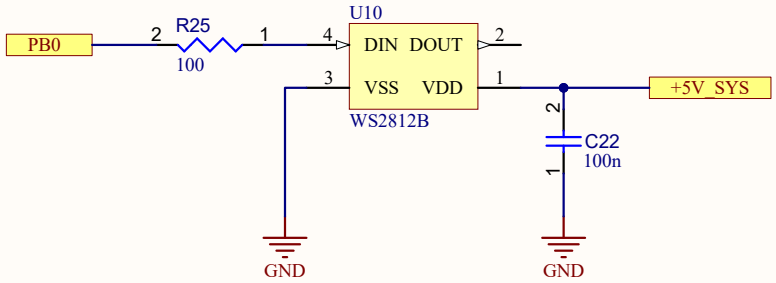
Title		
Size A	Number	Revision
Date: 9/10/2026		Sheet of
File: C:\Users\..\MCU Core(OLD).SchDoc		Drawn By:

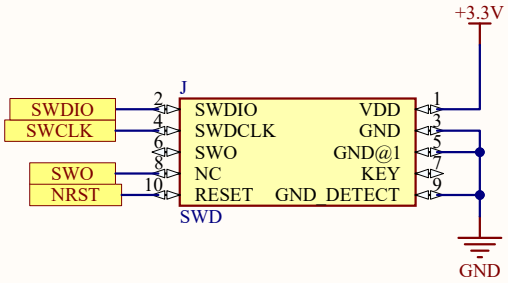

+5V_BATT
U3
VIN
VOUT
VOUT
Ground
AMS1117-3.3
C_LDO_IN
10uF
C_LDO_OUT
22uF
+3.3V
GND
C_VDD1
100n
C_VDD2
100n
C_VDD3
100n
C_VDD4
100n
C_BULK
2.2u
FB2
600 Ohm
C_VDDA1
1uF
C_VDDA2
10nF
R_NRST
10k
C_NRST
100n
SWD
NRST
BOOT0
VDD
VDDA
VSS
VCAP1
VCAP2
PH0/OSC_IN
PH1/OSC_OUT
PB8
PB9
I2C1_SCL
I2C1_SDA
STM32F446RET6
Y1
XRCGB24M000F3M00R0
NC
C8
15pF
C7
15pF
R_BOOT
10k
V_CAP1
V_CAP2
Title
Size
A
Number
Revision
Date:
9/10/2026
Sheet of
File:
C:\Users\..\MCU, 3.3V LDO & SUPPORT CIRCUITS.SchDoc



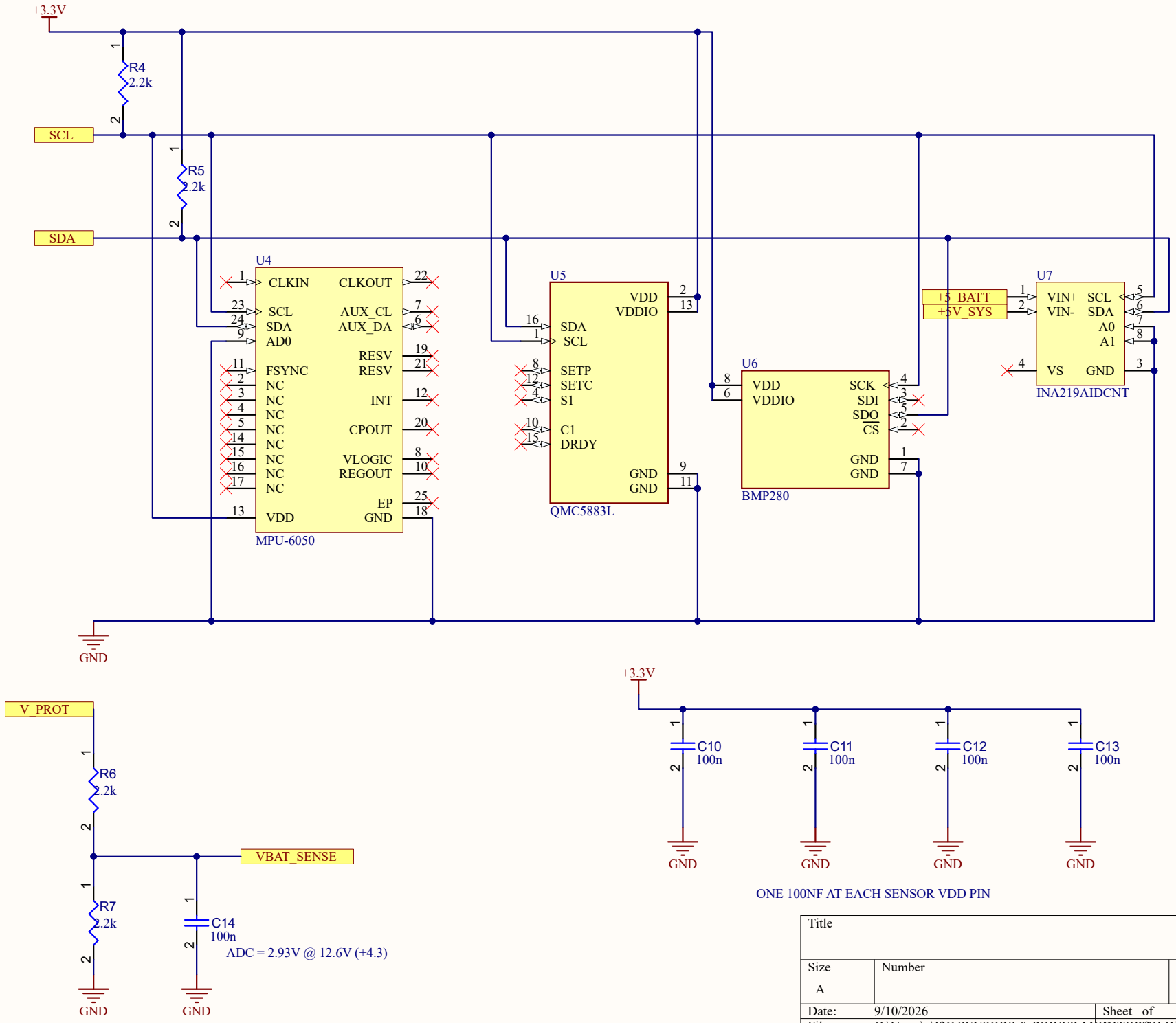
NEED LORA HERE

NEO 6M GPS





Title		
Size: A	Number	Revision
Date: 9/10/2026		Sheet of
File: C:\Users\..\RC, TELEMETRY, GPS & PERIPHERALS(old).SchDoc		Drawn By:


+3.3V
R4
2.2k
SCL
R5
2.2k
SDA
U4
CLKIN
CLKOUT
SCL
AUX_CL
SDA
AUX_DA
AD0
RESV
FSYNC
RESV
NC
INT
CPOUT
VLOGIC
REGOUT
EP
VDD
GND
MPU-6050
U5
VDD
VDDIO
SDA
SCL
SETP
SETC
S1
C1
DRDY
GND
GND
QMC5883L
U6
VDD
VDDIO
SCK
SDI
SDO
CS
GND
GND
BMP280
+5_BATT
+5V_SYS
U7
VIN+
VIN-
SCL
SDA
A0
A1
VS
GND
INA219AIDCNT
GND
V_PROT
R6
2.2k
VBAT_SENSE
R7
2.2k
C14
100n
ADC = 2.93V @ 12.6V (+4.3)
GND
GND
+3.3V
C10
100n
C11
100n
C12
100n
C13
100n
GND
GND
GND
GND
ONE 100NF AT EACH SENSOR VDD PIN
Title
Size
A
Number
Revision
Date: 9/10/2026
Sheet of
File: C:\Users\..\I2C SENSORS & POWER MONITORING(OLD).SchDoc
Drawn By: